TOTAL ANISOTROPY ON THE MOMENT CURVE FOR SIMPLICIAL 1-CYCLES

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ABSTRACT. We study total anisotropy for the Gorensteinification of the face ring of a simplicial cycle when the Artinian reduction is defined by the moment curve. Over every field and for every simplicial 1-cycle with arbitrary nonzero coefficients on its support, we prove that the middle multiplication form over the rational function field in the vertex parameters is anisotropic. The form is identified with the nonsingular quotient of a weighted Dirichlet form on the cut space. Its radical is dual to that of a linear configuration pencil on the cycle space. A nonzero maximal minor of this pencil has degree equal to its rank, forcing an edge contraction that preserves the rank. At such an edge, the associated one-parameter collision has rank jump one: its even residue is the contracted middle form and its odd residue is the one-dimensional form $\langle -\mu_e \rangle$. A discrete-valuation anisotropy argument and induction on the number of support vertices complete the proof.

1. INTRODUCTION

Let μ be a simplicial $(d-1)$-cycle over a field k, and let $\mathcal{B}^*(\mu)$ denote the Poincaré-duality algebra obtained by Gorensteinifying an Artinian reduction of the face ring of its support. The moment-curve anisotropy conjecture asks for a transcendental extension and an Artinian reduction for which every nonzero homogeneous $u \in \mathcal{B}^k(\mu)$, $k \le d/2$, has $u^2 \ne 0$. In the strong moment-curve version the extension is $k(t_v : v \in V)$ and the system of parameters is

$$\theta_i = \sum_{v \in V} t_v^i x_v, \qquad 1 \le i \le d.$$

This explicit version appears as Conjecture 6.1 in [1]; its displayed Theorem 1.1 establishes the corresponding p-power statement in characteristic p under a link hypothesis.

For one-dimensional simplicial spheres, hence for polygon supports, Papadakis and Petrotou proved generic anisotropy over every field [4, Theorem 1.3]. In characteristic two, the existence version for arbitrary simplicial cycles follows from [2, Theorem IV]. That condition-free result is stated for an appropriate Artinian reduction rather than explicitly for the same

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moment-curve parameters. We are not aware of a previous proof of the strong moment-curve statement over every field for arbitrary simplicial 1-cycles, whose supports may have several cycle-space directions, cut vertices, or several connected components, and whose cycle coefficients are arbitrary subject to the boundary equation.

Our main result proves the explicit moment-curve statement in that generality.

Theorem 1.1 (Main theorem)**.** *Let k be any field, and let μ be any simplicial* 1*-cycle with arbitrary nonzero coefficients on its support satisfying the simplicial cycle equations. Then the moment-curve middle form on $\mathcal{B}^1(\mu)$ over $K = k(t_v : v \in V)$ is anisotropic. Consequently the moment-curve Gorensteinification $\mathcal{B}^*(\mu)$ is totally anisotropic.*

A Lean 4 formalization of the graph-form theorem, the collision and induction arguments, and their transport to the actual degree-one and socle-degree-two low-degree interfaces accompanies this paper [5]. The formalized theorem has no characteristic hypothesis.

The proof has five steps. First, the middle form is written as a weighted Dirichlet form on the cut space. Second, its radical is identified with the radical of a configuration pencil on the cycle space. Third, a degree argument forces an edge on whose contraction the configuration rank is preserved. Fourth, the collision of the two endpoint parameters has exactly one positive valuation layer: the contracted form in even parity and a nonzero line in odd parity. Finally, a discrete-valuation argument and strong induction on the number of support vertices prove anisotropy.

The argument is independent of any finite graph classification. In particular, an edge is selected by polynomial divisibility, not by checking a list of supports or evaluating at points of the base field.

2. Simplicial 1-cycles and the middle form

Fix a finite simple graph $G = (V, E)$ and orient every edge. Write an oriented edge as $e = i \to j$ and define the incidence map

$$\partial : k^E \longrightarrow k^V, \qquad \partial(\mathbf{e}_e) = \mathbf{e}_j - \mathbf{e}_i.$$

A *simplicial* 1*-cycle with support G* is an element

$$\mu = \sum_{e \in E} \mu_e[e] \in \ker \partial$$

with $\mu_e \neq 0$ for every $e \in E$. Thus G is exactly the support of μ. Reversing the orientation of e replaces both $[e]$ and μ_e by their negatives and changes none of the forms below.

Let

$$K = k(t_v : v \in V), \qquad s_e = t_j - t_i \quad (e = i \to j).$$

Let $K[G]$ be the Stanley–Reisner ring of the one-dimensional simplicial complex G, and set

$$\theta_1 = \sum_{v \in V} t_v x_v, \qquad \theta_2 = \sum_{v \in V} t_v^2 x_v,$$
$$\mathcal{A}^*(G) = K[G]/(\theta_1, \theta_2).$$

The cycle gives a degree functional in degree two, normalized on an oriented edge $e = i \to j$ by

$$\deg(x_i x_j) = \frac{\mu_e}{t_i t_j (t_j - t_i)}. \tag{1}$$

The Gorensteinification is

$$\mathcal{B}^*(\mu) = \mathcal{A}^*(G)/\operatorname{ann}(\deg),$$

where the annihilator is taken with respect to multiplication followed by deg. It is a Poincaré-duality algebra of socle degree two. The middle form is

$$q_\mu(u, u') = \deg(uu') \qquad (u, u' \in \mathcal{B}^1(\mu)).$$

For a symmetric bilinear form b, write $b[x] = b(x, x)$ and call b *anisotropic* if $b[x] \neq 0$ for every nonzero x. This definition is used in every characteristic, including characteristic two.

We next derive the cut-space model directly. For $e = i \to j$, define the K-algebra map on the polynomial ring supported on e by

$$\sigma_e(x_i) = \frac{1}{t_i(t_i - t_j)}, \qquad \sigma_e(x_j) = \frac{1}{t_j(t_j - t_i)}, \qquad \sigma_e(x_v) = 0 \quad (v \notin \{i, j\}).$$

The functional

$$\Lambda(z) = -\sum_{e \in E} \mu_e s_e \, \sigma_e(z), \qquad z \in K[G]_2, \tag{2}$$

is the degree functional. Indeed, $\sigma_e(\theta_1) = 0$ and $\sigma_e(\theta_2) = 1$. Thus Λ kills $\theta_1 K[G]_1$, while

$$\Lambda(\theta_2 x_v) = \frac{1}{t_v}\left(\sum_{e \text{ out of } v} \mu_e - \sum_{e \text{ into } v} \mu_e \right) = 0$$

by $\partial\mu = 0$. On $x_i x_j$ it agrees with equation (1), and the edge monomials span the top degree after reduction. Formula (2) is the $d = 2$ moment-curve specialization of Lee's degree formula [3, Theorem 11].

For $u = \sum_v c_v x_v$ and $u' = \sum_v c'_v x_v$, put $z_v = c_v/t_v$ and $z'_v = c'_v/t_v$. Then

$$\sigma_e(u) = \frac{z_j - z_i}{s_e},$$

and, since σ_e is an algebra homomorphism, applying equation (2) directly to uu' gives

$$q_\mu(z, z') = -\sum_{e = i \to j} \frac{\mu_e}{s_e} (z_j - z_i)(z'_j - z'_i). \tag{3}$$

Let $\delta = \partial^{\mathsf{T}}$ and define the cycle and cut spaces

$$\mathcal{Z} = \ker(\partial : K^E \to K^V), \qquad \mathcal{C} = \operatorname{im}(\delta : K^V \to K^E).$$

On $\mathcal{C}$ define

$$\beta_t(c, c') = -\sum_{e \in E} \frac{\mu_e}{s_e} c_e c'_e. \tag{4}$$

Since $\delta z = (z_j - z_i)_e$, equation (3) is the pullback of β_t. The two parameter vectors become $\mathbf{1}$ and t in the z-coordinates, and $\delta t = s = (s_e)_e$. Moreover,

$$\beta_t(s, \delta z) = -\langle \mu, \delta z \rangle = -\langle \partial\mu, z \rangle = 0. \tag{5}$$

Proposition 2.1 (Cut-space model)**.** *There is an isometry*

$$(\mathcal{B}^1(\mu), q_\mu) \cong \big(\mathcal{C}/\operatorname{rad}(\beta_t|_{\mathcal{C}}), \overline{\beta}_t\big).$$

This remains valid when G is disconnected.

Proof. The map $z \mapsto \delta z$ takes $K^V/\langle \mathbf{1}, t\rangle$ onto $\mathcal{C}/Ks$. For a disconnected graph its additional kernel consists of potentials constant on each component. These map to zero and therefore lie in the multiplication radical. By equation (5), quotienting by Ks does not alter the eventual nonsingular quotient. Finally, in socle degree two the degree-one part of ann(deg) is precisely the radical of the middle pairing. Removing it gives the stated isometry. □

3. The configuration form and radical duality

Put

$$D(t) = \operatorname{diag}\left(\frac{s_e}{\mu_e} : e \in E\right).$$

On the cycle space define the configuration form

$$\gamma_t(\eta, \nu) = \sum_{e \in E} \frac{s_e}{\mu_e} \eta_e \nu_e = \langle D(t)\eta, \nu \rangle. \tag{6}$$

The dot product identifies $\mathcal{Z}^\perp = \mathcal{C}$. Since $D(t)\mu = s$,

$$\gamma_t(\mu, \nu) = \langle s, \nu \rangle = \langle t, \partial\nu \rangle = 0,$$

so $K\mu \subseteq \operatorname{rad}(\gamma_t|_{\mathcal{Z}})$.

Proposition 3.1 (Radical duality)**.** *Multiplication by $D(t)$ induces mutually inverse isomorphisms*

$$\operatorname{rad}(\gamma_t|_{\mathcal{Z}}) \underset{D(t)^{-1}}{\overset{D(t)}{\longleftrightarrow}} \operatorname{rad}(\beta_t|_{\mathcal{C}}).$$

Under this correspondence μ maps to s.

Proof. Suppose $\eta \in \operatorname{rad}(\gamma_t|_{\mathcal{Z}})$. Then $D(t)\eta \in \mathcal{Z}^\perp = \mathcal{C}$. For $c' \in \mathcal{C}$,

$$\beta_t(D(t)\eta, c') = -\langle \eta, c' \rangle = 0,$$

so $D(t)\eta \in \operatorname{rad}(\beta_t|_{\mathcal{C}})$. Conversely, if $c \in \operatorname{rad}(\beta_t|_{\mathcal{C}})$ and $\eta = D(t)^{-1}c$, then $\langle \eta, c' \rangle = 0$ for every $c' \in \mathcal{C}$, hence $\eta \in \mathcal{Z}$. Also $D(t)\eta = c \in \mathcal{C} = \mathcal{Z}^\perp$, so $\gamma_t(\eta, \nu) = 0$ for every $\nu \in \mathcal{Z}$. □

Assume now that G is connected, with $n = |V|$ and $m = |E|$. Put $g = \dim \mathcal{Z} = m - n + 1$. Fix once and for all a k-linear complement

$$\ker(\partial : k^E \to k^V) = k\mu \oplus Y. \tag{7}$$

In a fixed k-basis of Y, the restriction of γ_t has a matrix $A(t)$ whose entries are homogeneous linear polynomials in the t_v. Let

$$\rho = \operatorname{rank}_K A(t), \qquad a = (g-1) - \rho. \tag{8}$$

Then $\dim \operatorname{rad}(\gamma_t|_{\mathcal{Z}}) = 1 + a$. By proposition 3.1, the cut form has the same radical dimension and therefore rank

$$\operatorname{rank}(\beta_t|_{\mathcal{C}}) = (n-1) - (1+a) = n - 2 - a. \tag{9}$$

All specializations below use the fixed complement Y; no parameter-dependent complement is introduced.

4. Contraction through a labelled multigraph

Fix an oriented edge $e = u \to v$. Let G/e_{multi} be obtained by identifying u and v, deleting the loop e, and retaining every other edge as a separately labelled multiedge. Orient retained edges compatibly, changing both their coordinate and coefficient signs when necessary.

Proposition 4.1 (Exact contraction)**.** *Deleting the e-coordinate induces an isomorphism*

$$p_e : \mathcal{Z}(G) \xrightarrow{\cong} \mathcal{Z}(G/e_{\text{multi}}).$$

Under this isomorphism, specialization at $s_e = t_v - t_u = 0$ identifies the configuration form of G exactly with the configuration form of the labelled contracted multigraph.

Proof. Adding the boundary equations at u and v shows that deleting the e-coordinate sends a circulation to a circulation on the contraction. Its kernel would be a circulation supported only on the nonloop edge e, hence is zero. Both cycle spaces have dimension

$$(m-1) - (n-1) + 1 = m - n + 1 = g,$$

so the map is onto. After imposing $t_v = t_u$, the term $(s_e/\mu_e)\eta_e\nu_e$ vanishes, and every retained term becomes exactly the corresponding term of the contracted labelled multigraph. This proves the form identity. $\square$

Only after this cycle-space identification do we aggregate parallel edges. Give parallel retained edges a common orientation. In the potential form their contributions have the same moment difference and the same potential difference, so

$$-\frac{\mu_f}{s_f}c_f^2 - \frac{\mu_{f'}}{s_f}c_f^2 = -\frac{\mu_f + \mu_{f'}}{s_f}c_f^2.$$

Thus they may be replaced by one simple edge with signed coefficient $\mu_f + \mu_{f'}$. A zero aggregate is deleted. The resulting chain, denoted $\overline{\mu}_e$, is a simplicial 1-cycle, possibly zero and possibly with disconnected support or isolated inactive vertices. Its potential form is identical to that of the contracted labelled multigraph. We make no claim that their cycle spaces are isomorphic after aggregation.

5. A rank-preserving edge forced by degree

Lemma 5.1 (Rank-preserving edge)**.** *Let G be connected and simple. With $A(t)$ and ρ as above, at least $m-\rho$ edges $e \in E$ satisfy*

$$\operatorname{rank} A(t)\big|_{s_e=0} = \rho.$$

In particular, at least one such edge exists.

Proof. If $\rho = 0$, use the empty minor $P = 1$; no specialization can lower the rank. Assume $\rho > 0$ and choose a nonzero $\rho \times \rho$ minor $P(t)$ of $A(t)$. Because the entries of $A(t)$ are homogeneous linear polynomials, P is homogeneous of degree ρ.

Every entry depends only on moment differences. Fixing one vertex parameter to zero therefore realizes P as a nonzero polynomial in a UFD on the remaining $n-1$ parameters. Since G is simple, the edge differences s_f are pairwise nonassociate irreducible linear forms. If the rank drops after imposing $s_f = 0$, then every ρ-minor maps to zero in the domain $k[t]/(s_f)$; in particular, s_f divides P. Pairwise nonassociate irreducibles corresponding to rank-dropping edges therefore have product dividing P. Their number is at most $\deg P = \rho$.

Finally,

$$\rho \le \dim Y = g - 1 = m - n < m.$$

Hence at least $m - \rho > 0$ edges preserve rank. The argument is polynomial divisibility, not evaluation on k-rational points; no infinitude or perfection hypothesis on k is used. □

6. A one-rank discrete-valuation collision

We isolate the linear algebra needed at a collision. A symmetric bilinear form may be degenerate; its *nonsingular quotient* is the quotient by its radical. We write $\mathrm{NS}(q)$ for the nonsingular quotient of a possibly degenerate form q.

Lemma 6.1 (One-rank DVR lemma)**.** *Let R be a discrete valuation ring with uniformizer π, fraction field L, and residue field F. Let $M \in \mathrm{Sym}_d(R)$ and assume*

$$\operatorname{rank}_L M = \operatorname{rank}_F \overline{M} + 1. \tag{10}$$

Suppose $\overline{r} \in \ker \overline{M}$ has an integral lift $\widetilde{r} \in R^d$ such that

$$\pi^{-1} M(\widetilde{r}, \widetilde{r}) \bmod \pi \ne 0. \tag{11}$$

Then the nonsingular quotient of M_L is an orthogonal sum of a unimodular lift of the nonsingular quotient of $\overline{M}$ and one line of the form $\pi\langle u\rangle$, where

$$\overline{u} = \pi^{-1} M(\widetilde{r}, \widetilde{r}) \bmod \pi.$$

Every remaining direction is an exact zero block over L. The class in equation (11) *is independent of the integral lift.*

Proof. If $\widetilde{r}$ is replaced by $\widetilde{r} + \pi h$, its norm changes by

$$2\pi M(\widetilde{r}, h) + \pi^2 M(h, h).$$

Because $\overline{r} \in \ker \overline{M}$, one has $M(\widetilde{r}, h) \in \pi R$, so both terms are divisible by π^2 in every characteristic. This proves lift independence.

Lift a complement on which $\overline{M}$ is nonsingular. Its Gram block is unimodular, so integral block elimination splits it off; this only inverts the Gram block. On the remaining module the Schur complement S is divisible by π. The Schur correction to the norm of $\widetilde{r}$ begins in order π^2, so the reduction of $T = \pi^{-1} S$ has a vector of nonzero norm. Lift it to a vector w for which $T(w, w)$ is a unit. The projection

$$x \longmapsto T(x, w)T(w, w)^{-1} w$$

splits Rw orthogonally. This uses no division by 2.

The rank identity equation (10) says that S_L has rank one. Consequently the orthogonal complement of the split line is identically zero over L. This gives the claimed unimodular block, the single $\pi\langle u\rangle$ block, and the exact zero block. $\square$

We also record the elementary valuation argument that separates the two collision parities.

Lemma 6.2 (Parity separation)**.** *Let F be any field. For $i = 0, 1$, let*

$$Q_i \in \mathrm{Sym}_{d_i}(F[[\pi]]).$$

If the reductions $\overline{Q}_0$ and $\overline{Q}_1$ are anisotropic over F, then $Q_0 \perp \pi Q_1$ is anisotropic over $F((\pi))$. Moreover, if q is an anisotropic symmetric bilinear form over F, then it remains anisotropic over the purely transcendental extension $F(T)$.

Proof. For a nonzero Laurent-series vector x in the ith block, let m_i be the minimum valuation of its coordinates. After writing $x = \pi^{m_i} x_0$, the vector x_0 is integral and has nonzero reduction. Anisotropy of $\overline{Q}_i$ makes $Q_i[x_0]$ a unit, and therefore

$$\mathrm{val}(Q_i[x]) = 2m_i.$$

Thus every nonzero value represented by Q_0 has even valuation, whereas every nonzero value represented by πQ_1 has odd valuation. If both block vectors are nonzero, their values cannot cancel; if only one is nonzero, its value is nonzero. This proves the first statement without a characteristic assumption and allows arbitrary integral lifts of the two residue forms.

For the second statement, embed $F(T)$ into $F((T))$. If a zero existed, scale its coordinates so that they are integral and at least one has nonzero reduction. Reducing modulo T would give a nonzero zero of q over F, a contradiction. $\square$

7. The collision calculation

Let $e = u \to v$ be a rank-preserving edge from lemma 5.1, and put

$$\pi = t_v - t_u, \qquad F = k(t_w : w \in V \setminus \{v\}).$$

Thus the original moment field is $F(\pi)$ via $t_v = t_u + \pi$. Use potential coordinates and impose the double gauge

$$z_u = z_v = 0.$$

For $\pi \neq 0$, this subspace is complementary to the two affine radical directions $\mathbf{1}$ and t. The edge e contributes zero on this gauge. Every other denominator is a unit at $\pi = 0$, so the restricted potential form has a matrix

$$M(\pi) \in \mathrm{Sym}_{n-2}(F[[\pi]]).$$

Recall $a = (g-1) - \rho$. By equation (9),

$$\operatorname{rank}_{F((\pi))} M(\pi) = n - 2 - a. \tag{12}$$

At $\pi = 0$, exact contraction gives a connected labelled multigraph on $n-1$ vertices with the same cycle-space dimension g. Rank preservation gives configuration rank ρ, hence radical dimension $g - \rho = 1 + a$. Its cut space has dimension $n-2$, so

$$\operatorname{rank}_F M(0) = n - 3 - a. \tag{13}$$

Thus the generic rank exceeds the special rank by exactly one.

The contracted slope, in the double gauge, is

$$r_w = t_w - t_u \qquad (w \in V \setminus \{u, v\}). \tag{14}$$

It lies in $\ker M(0)$. We now compute its first divided norm with all signs visible. Keep $e = u \to v$ and orient every other edge incident with v as $v \to w$, changing its coefficient sign if necessary. Write $a_w = t_w - t_u$. The term of such an edge in the value of the potential form on r is

$$-\frac{\mu_{vw}}{t_w - t_v} a_w^2 = -\frac{\mu_{vw}}{a_w - \pi} a_w^2 = -\mu_{vw} a_w - \pi \mu_{vw} + O(\pi^2).$$

All terms not incident with v are independent of π, and the term of e vanishes because $z_u = z_v = 0$. Since $M(0)[r, r] = 0$, the coefficient of π is therefore

$$\pi^{-1} M(\pi)[r, r] \bmod \pi = -\sum_{v \to w,\, w \neq u} \mu_{vw}.$$

The coefficient of v in $\partial\mu = 0$ is

$$\mu_e - \sum_{v \to w,\, w \neq u} \mu_{vw} = 0.$$

Consequently

$$\pi^{-1} M(\pi)[r, r] \bmod \pi = -\mu_e \neq 0. \tag{15}$$

Applying lemma 6.1 to equations (12), (13) and (15) gives the exact collision decomposition

$$\mathrm{NS}(M(\pi)) \cong Q_{\mathrm{contr}} \perp \pi\langle -\mu_e \rangle \quad \text{over } F((\pi)), \tag{16}$$

where Q_{contr} is a unimodular lift of the nonsingular middle form of the aggregated contracted cycle $\overline{\mu}_e$. Every pre-existing generic radical direction lies in the exact zero block. There are no higher nonzero collision layers.

8. Induction on support vertices

We first separate disconnected components. The next lemma is needed because an orthogonal sum of anisotropic forms need not be anisotropic over an arbitrary common field.

Lemma 8.1 (Component scale separation). *Let $\mu = \mu_1 + \cdots + \mu_c$ be the decomposition of a simplicial 1-cycle over its connected support components. If every component middle form is anisotropic over its own moment field, then the middle form of μ is anisotropic over the full moment field.*

Proof. The cut space, its weighted form, and its radical are orthogonal direct sums over the components, so the global nonsingular middle form is the orthogonal sum of the component forms.

Introduce independent variables τ_v and $y_2, \ldots, y_c$, put $y_1 = 1$, and embed the full moment field by

$$t_v \longmapsto y_i \tau_v \qquad (v \in V(\mu_i)).$$

The images of the t_v are algebraically independent. Under this embedding, every moment difference in component i acquires the factor y_i, so its middle form acquires the scalar y_i^{-1}. Over

$$L = k(\tau_v : v \in V)((y_2)) \cdots ((y_c))$$

the pulled-back form is

$$q_1 \perp y_2^{-1} q_2 \perp \cdots \perp y_c^{-1} q_c.$$

By lemma 6.2, adjoining the unused τ-variables preserves anisotropy of every q_i. Applying the one-variable parity argument successively in $y_2, \ldots, y_c$ shows that the displayed sum is anisotropic over L. An isotropic vector over the original moment field would remain isotropic after this embedding, which is impossible. □

Proof of theorem 1.1. We use strong induction on the number of active support vertices. Suppose the theorem holds for every simplicial 1-cycle with fewer than n active vertices, and let μ have n active vertices.

If the support is disconnected, every connected component has fewer than n vertices. The induction hypothesis and lemma 8.1 prove the result.

Assume therefore that the support is connected. The rank-preserving edge lemma applies without any 2-connectedness assumption, so cut vertices require no separate reduction. Choose a rank-preserving edge e and form the aggregated contracted cycle $\overline{\mu}_e$. Delete zero aggregate edges and isolated inactive vertices. If the result is zero, its nonsingular middle form is the zero-dimensional form, which is anisotropic by convention. Otherwise it has at most $n-1$ active vertices, even if its support is disconnected, so its middle form is anisotropic over its own moment field by induction.

The residue field F used in the collision may contain moment variables of vertices absent from the active contracted support. These are purely transcendental variables, and anisotropy survives adjoining them by lemma 6.2. Hence the reduction $\overline{Q}_{\mathrm{contr}}$ of the unimodular block in equation (16) is anisotropic over F. The odd residue $\langle -\mu_e \rangle$ is a nonzero one-dimensional form and is also anisotropic. Since both residue forms are anisotropic, the integral parity-separation lemma makes the original nonsingular middle form anisotropic over $F((\pi))$. It is therefore anisotropic over the subfield $F(\pi)$: an isotropic vector over the subfield would remain isotropic after extension. Finally $F(\pi) = k(t_v : v \in V)$ under $t_v = t_u + \pi$. This completes the induction and proves anisotropy of $\mathcal{B}^1(\mu)$.

Since $\mathcal{B}^*(\mu)$ has socle degree two, the only positive degree at most half the socle degree is degree one. Degree-zero nonzero elements have nonzero squares. Thus middle anisotropy is exactly total anisotropy in this case. □

9. Characteristic two

The proof is uniform in the characteristic. In particular, equation (3) was derived by evaluating the bilinear expression uu', not by polarizing a quadratic expression. The DVR proof uses unimodular block elimination and the displayed orthogonal projection, neither of which divides by 2. Finally, lemma 6.2 uses only valuation parity and residue anisotropy. These observations are exactly what permits the argument to work in characteristic two, where the diagonal map $x \mapsto q(x, x)$ is quasilinear.

Prior work already supplies the literal existence statement in characteristic two: Theorem IV of Adiprasito–Papadakis–Petrotou [2], specialized to $p = 2$, gives an anisotropic Artinian reduction for every simplicial cycle over an appropriate transcendental extension. The displayed Theorem 1.1 of [1] gives the explicit moment-curve reduction in characteristic two under its condition $(*)$ and points to [2] for removing that link condition. The condition-free theorem is stated for an appropriate Artinian reduction rather than explicitly for the same moment-curve parameters. Thus theorem 1.1 supplies, in dimension two, the condition-free strong moment-curve statement in characteristic two as well as in every other characteristic.

10. Concluding remark on scope

The result extends the one-dimensional theory from simplicial 1-spheres to arbitrary simplicial 1-cycles with nonzero coefficients on their support satisfying the cycle equations. The essential mechanism is not a classification of graph supports: it is the degree bound on a maximal minor of the cycle-space configuration pencil, followed by the exact one-rank collision at a rank-preserving contraction. No higher-dimensional analogue is asserted here.

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